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Effects of prosocial cartoon models on aggressive cognitions and aggressive behaviors

Qian Zhang[a,b,*], Yi Cao[b,c], JingJin Tian[b,d], Kim El-Lim[e], Douglas A. Gentile[e]

[a] Center for Studies of Education and Psychology of Minorities in Southwest China, Southwest University, Chongqing, China
[b] Faculty of Education, Southwest University, Chongqing, China
[c] The Third Kindergarten in Chengdu, Sichuan, China
[d] The Dawn Innovation Kindergarten, Chongqing, China
[e] Department of Psychology, Iowa State University, Ames, IA, USA

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ABSTRACT

Although research has examined violent cartoon effects, little research has evaluated the effects of prosocial cartoons on kindergartners. Based on the general learning model (GLM) and social learning model (SLM), we examined how prosocial cartoon models affect aggressive cognitions and aggressive behaviors. A total of 174 children aged 5 and 6 years (50% girls) who had been nominated by their teacher and peers as aggressive were recruited from three Chinese kindergartens. Half of them were assigned to the experimental group to watch prosocial cartoons for five consecutive days, while the other half were assigned to the control group that did not watch cartoons. The modified semantic classification task (MSCT) and the competitive reaction time task (CRTT) were employed to measure aggressive cognitions and aggressive behaviors. Children who watched the prosocial cartoons displayed lower aggressive cognitions and aggressive behaviors than those who did not watch the cartoons. Specifically, watching prosocial cartoons reduced aggressive cognitions and aggressive behaviors more for boys than girls. Aggressive cognitions partially mediated the effect of prosocial cartoon viewing on aggressive behaviors.

1. Introduction

Prosocial behaviors can be defined at several levels (Penner, Dovidio, Piliavin, & Schroeder, 2005). Indeed, children demonstrate prosocial behaviors in many ways, including helping, donating, sharing, comforting, and cooperating (Drummond, Waugh, Hammond, & Brownell, 2015). The core aspect underlying different morphological behaviors is an intention to help others, and this definition can also be applied to media characters. Accordingly, we conceptualize prosocial cartoon models as cartoon figures who help others to solve difficulties, including helping, donating, sharing, comforting, and cooperating.

This definition is largely parallel to the standard definition of aggression—any behavior directed toward another individual that is carried out with the intent to cause harm (Anderson & Bushman, 2002a). Preschoolers' aggressive behaviors can harm not only peer relationships (Crick et al., 2006), but also teacher-student relationships (Büyüktaskapu, Devlet, & Hayriye, 2017). There are many mechanisms through which viewing violent media can influence subsequent aggressive behaviors. One well-established mechanism is through priming cognitive networks of aggressive schemas (Berkowitz, 1990); that is, aggressive cognitions can elicit aggressive behaviors. Because aggressive and prosocial motivations generally oppose each other, we assume that aggressive cognitions will also be an important mediator between prosocial media and prosocial behavior, although there is far less research on this than for violent media.

1.1. General learning model and social learning model

Several theories have been proposed to explain relations between media exposure and behavior, including the general learning model (GLM) and social learning model (SLM). The GLM posits that prosocial media can increase prosocial cognitions and behaviors. This is a central assumption of most modern learning theories: People learn from things they see. Alternatively, children's aggression can also be inhibited in both short-term and long-term when they are exposed to prosocial media. Situational factors (e.g., media content, duration of exposure)

* Corresponding author at: Faculty of Education, Southwest University, Tiansheng Rd. 2, Beibei District, Chongqing 400715, China.
E-mail addresses: zhq@swu.edu.cn (Q. Zhang), 2271613535@qq.com (Y. Cao), 9114978@qq.com (J. Tian), ellimkim@iastate.edu (K. El-Lim), dgentile@iastate.edu (D.A. Gentile).

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and personal factors (e.g., sex, age) can interact to affect individual's learning states (e.g., cognition, emotion) and social behaviors (Buckley & Anderson, 2006; Gentile, Groves, & Gentile, 2014; Maier & Gentile, 2012). Importantly, each exposure is a learning trial that, if repeated, can lead to well-rehearsed knowledge structures. The SLM notes that children acquire aggressive and/or prosocial behaviors not only through imitation of models in real life (e.g., parents), but also from models in cartoons and other media contents (Bandura, 1973; Bandura, Ross, & Ross, 1961). Thus, role models in cartoons may function as a potential contributor to children's acquisition of aggressive and/or prosocial behavior. Based on these two theories, we aim to test the interactive effect of prosocial cartoon models (situational factor) and gender (personal factor) on children's aggression.

1.2. Media and children's aggression

Bandura and his colleagues demonstrated that children can learn aggressive behavioral patterns simply from observation (Bandura et al., 1961). Hundreds of studies explored how children learn from media, and what they learn from it (e.g., Anderson & Bushman, 2002a; Anderson et al., 2010; Bushman & Huesmann, 2006). Despite some scholars arguing that the effect size of media effect is too small to be practical (e.g., Ferguson, 2007), the accumulated evidence since the 1960s shows that children and adults can learn aggressive attitudes, beliefs, and behaviors from the media (e.g., Anderson & Bushman, 2002b; Gentile, 2014).

Although there are many aspects of how media can affect viewers, the core of the effect relies on learning. Humans learn in many ways, and observational learning is one of such important learning mechanisms (e.g., Bandura, 1973). Social learning theory proposes that people can learn not only behaviors themselves but also preferences, attitudes, cognitions, feelings, perceptual schemata and social norms simply from watching others' behaviors and the context and consequences of those behaviors. At present, there are only a handful of research studies on this topic because most research has focused on the effect of violent media. The basic learning effect from media has been well-established, particularly with violent media (e.g., Gentile, 2014). What has been less studied are the variables that moderate and/or mediate media effects. What are the mechanisms underlying these processes, and can they be less direct? For example, rather than prosocial media increasing prosocial cognitions and behavior, can prosocial media reduce aggressive cognitions and behavior? The present study focuses on this set of questions.

There has been some debates whether prosocial media and aggression-related variables are correlated: Some researchers have suggested that prosocial media could reduce aggression-related variables such as aggressive cognitions, aggressive affect, hostile attribution bias, and aggressive behaviors (Coyne, Padilla-Walker, Holmgren, Davis, & Hawkins, 2017; Greitemeyer & Osswald, 2009; Kamper-Demarco & Ostrov, 2018), whereas other researchers were skeptical that prosocial media could reduce aggression-related variables (e.g., Sestir & Bartholow, 2010). To remedy this controversy, it is necessary to test whether prosocial media can indeed decrease aggression-related variables. Furthermore, it is important to replicate prosocial media research in a non-individualist East Asian sample for generalizability.

1.3. Cartoons and aggression

At this time, cartoon violence effects on aggression is still, surprisingly, a controversial topic. Most researchers have found positive correlations between cartoon violence and aggression (e.g., Kirsh, 2006), although some researchers proposed an opposing view (Blumberg, Bierwirth, & Schwartz, 2008). Most of this research has focused on cartoon violence, with comparatively little research testing the causal effect of cartoon prosociality on aggression. Children often regard cartoon characters as role models, and exposure to such cartoons can promote adaptive behaviors such as healthy eating while reducing problematic behaviors (e.g., Gonçalves et al., 2018). Children can also learn aggressive behaviors from watching cartoon characters (Soydan, Pirpir, & Azak, 2017). Following this logic, fostering prosocial behavior through cartoon characters can be a strategy to prevent children from engaging in risky behaviors (e.g., Centers for Disease Control and Prevention, 2009). Thus, we propose the first hypothesis (H1): *Watching prosocial cartoon models will diminish aggressive cognitions and aggressive behaviors.*

1.4. Gender and aggression

An abundance of studies found that boys display more physical/direct aggression and more attentional bias to aggressive stimuli than girls do during early childhood (Alink et al., 2006; Baillargeon et al., 2007; Björkqvist, 2018; Kung, 2018). Specifically, boys show more physical aggression than girls do after being exposed to violent television cartoons and games (Luther & Legg, 2010). Importantly, co-playing prosocial video games is related to resolving sibling conflicts more for girls than for boys (Coyne, Jensen, Smith, & Erickson, 2016), and primary school boys also exhibit more aggressive behaviors than girls do after a brief exposure to prosocial video games (Li, Niu, & Wen, 2016). Likewise, men demonstrate more aggressive thoughts than women do after hearing a song with lyrics that convey prosocial contents (Böhm, Ruth, & Schramm, 2016). Thus, we propose the second hypothesis (H2): *Boys will display more aggressive cognitions and aggressive behaviors than girls do after watching prosocial cartoon models.*

1.5. Aggressive cognition and aggressive behavior

Media violence exposure can increase aggressive thoughts, aggressive affect and physiological arousal (Anderson & Bushman, 2001), and these subsequently predict aggressive behaviors (Anderson et al., 2004; Armanda, Ferreira, & Haase, 2012; Gentile, Bender, & Anderson, 2016). Less research has examined how prosocial media exposure relates to aggressive cognitions. One study with young adults found that aggressive cognition mediated the prosocial video game effect on aggressive behavior such that playing a prosocial video game decreased aggressive cognition, which in turn reduced aggressive behavior (Greitemeyer, Agthe, Turner, & Gschwendtner, 2012). In this regard, aggressive cognition may also serve as a mediator between prosocial cartoon models and aggressive behaviors, although this has not been studied with children this young to our knowledge. Thus, we state the third hypothesis (H3): *Aggressive cognition will mediate the prosocial cartoon model effect on aggressive behavior.*

1.6. Relative lack of prosocial cartoon research on kindergartners

To the best of our knowledge, few researchers have examined the effects of prosocial cartoons on young children's aggression, despite studies showing that media-related factors (e.g. media violence) can be risk factors for aggression (e.g., Anderson et al., 2017). A majority of these media studies focused on adolescents aged between 12 and 19 years, casting ambiguity on whether media effect can be generalized to younger children such as kindergartners. With the rapid development of the animated cartoon industry, cartoons are undoubtedly an influential mass media source for children to observe and acquire social behaviors (Yoon & Malecki, 2010). Cartoons may be a double-edged sword for kindergartners' social behaviors due to poor cartoon content quality in the Chinese market, such as the inclusion of biased gender role depictions (Cui, 2016). Nonetheless, little research has been conducted to test the causal relations between prosocial cartoons and young children's aggression in China.

1.7. The present experiment

The goal of this research is to evaluate prosocial cartoon effects on aggressive cognitions and aggressive behaviors of young children. Due to ethical concerns, we used proxy measures to represent children's aggressive cognition and aggressive behavior in a laboratory setting. Specifically, we used a modified semantic classification task (MSCT; Bowers & Turner, 2003) where children reacted to aggression-related pictures as a measure of the accessibility of aggressive cognitions, and the competitive reaction time task (CRTT; Taylor, 1967) where children were able to deliver noise blasts to opponents as the measure of aggressive behavior. The main study questions are the following: (1) Do prosocial cartoons affect aggressive cognitions and aggressive behaviors? (2) Are these effects moderated by gender? (3) Is the effect of prosocial cartoons on aggressive behaviors mediated by aggressive cognitions?

2. Method

2.1. Participants

In the spring of 2019, we randomly recruited 945 five- to six-year-old children (435 boys, 510 girls) from three kindergartens in China. From the preliminary sample, we selected children with relatively high aggressive behavioral tendencies based on peer and teacher nominations, which are commonly used in studies with children (see Rodkin, Ryan, Jamison, & Wilson, 2012). Children are asked, for example, "Who do you think are the bullies in your class?" Lab assistants asked the children to list at least ten classmates in order. Using this information, we excluded 679 participants who were relatively low in aggressiveness. The remaining sample included 266 children nominated by their peers and teachers to be aggressive[1]. To match the similar number of gender under each condition, we invited 174 children (50% girls, 50% 5-year-olds) to participate in the experiment. Half of them were randomly assigned to either an experimental group to watch the prosocial cartoons or a control group to not watch the cartoons.

2.2. Experimental design

A 2 (prosocial cartoon: watching vs. not watching) X 2 (gender: boys vs. girls) design was employed. The independent variables included both situational and personal variables identified by the GLM (Buckley & Anderson, 2006). The situational variable was prosocial cartoon, and the personal variable was gender. The dependent variables were aggressive cognitions (reaction time to aggressive pictures) and aggressive behaviors (noise intensity setting for the opponents).

2.3. Materials

2.3.1. Prosocial cartoon models

Five cartoons with prosocial behavior models were used as the experimental materials. Cartoon 1 was an episode titled "Rusty to the rescue" from *Handy Manny* series (U.S./Canada). In this cartoon episode, the main character Manny and his magical tools rushed to the school's jungle gym to save the trapped child (*model for helping and cooperating*). Cartoon 2 comprised two episodes from *Peppa Pig* (United Kingdom): The first one is titled "Fun run" in which Daddy Pig (main character's father) does a fun run to raise money for repairing the roof of the kindergarten (*model for donation*), and the second is titled "Piggy in the middle", which shows ball-throwing games among the main character Peppa and her family (*model for cooperating and helping*). Cartoon 3 included an episode titled "Pups save the mayor's race" from *PAW Patrol* (U.S.). In this episode, a ten-year-old boy named Ryder organizes the PAW patrol team (comprised of dogs) to repair Mayor Goodway's boat, to prevent Mayor Humdinger from cheating, and to further help Mayor Goodway to win the race (*model for cooperating and helping*). Cartoon 4 was "New year" episode from *Boonie Bears* series (China). It tells a story of two bears named Briar and Bramble who help Vick, a human logger, to catch his train so that Vick could go home and celebrate the Chinese New Year (*model for helping, comforting, sharing and cooperating*). At last, cartoon 5 consisted of two episodes from the cartoon series *Big-Headed Kid and Small-Headed Father* (China): The first one, titled "A small animal protection team," tells a story of big-headed son and his small-headed father setting up a small animal protection team to save wild animals (*model for helping and cooperating*). In the second episode "A girl named Mao Rong Rong is homesick," the father and son help a girl to decorate her house like her hometown so that she can feel a sense of belonginess (*model for helping, sharing, comforting and cooperating*). All the cartoons were dubbed in Mandarin.

Based on previous research with media violence (e.g., Barlett, Branch, Rodeheffer, & Harris, 2009), we confined the duration of each cartoon to be 10 min or less. To ensure that the selected cartoons do include prosocial behavior models, we ran a pilot study to evaluate the cartoons on the following categories: (1) prosocial scene, (2) prosocial content, (3) interest/fun, (4) popularity, (5) excitement, and (6) pleasure. A total of 61 raters (38 undergraduate students, 22 graduate students, 1 child psychology professor; M_{age} = 21.98, SD = 2.74; female = 49, male = 12) were invited to rate the selected cartoons, using a 5-point Likert-type scale (1 = not at all, 5 = a lot). In this study, we postulated that cartoons with an average rating over 3 for all six categories would be considered as good prosocial materials. Results from a total of 61 sets of valid ratings were used (see Table 1).

Following prior research that rated media violence based on its violent scene and content (Anderson & Dill, 2000), we rated prosocial media based on its prosocial scene and content. As shown in Table 1, the rankings (from high to low) of the five cartoons were: (1) Handy Manny (M_{scene} = 4.07, SD = 0.65; $M_{content}$ = 4.07, SD = 0.70), (2) Big-Headed Kid and Small-Headed Father (M_{scene} = 3.90, SD = 0.70; $M_{content}$ = 3.93, SD = 0.70), (3) Boonie Bears (M_{scene} = 3.57, SD = 1.07; $M_{content}$ = 3.61, SD = 1.08), (4) Peppa Pig (M_{scene} = 3.39, SD = 0.78; $M_{content}$ = 3.52, SD = 0.70), and finally (5) PAW Patrol (M_{scene} = 3.36, SD = 0.90; $M_{content}$ = 3.39, SD = 0.86). As demonstrated, the average rating of each cartoon was more than 3 for both prosocial scene and content, leading to the conclusion that the five selected cartoons had prosocial models. In addition, rating scores of interest/fun, popularity, excitement, and pleasure categories were also over 3 points, indicating that these cartoon models were suitable for children to watch. Thus, we judged the five cartoons to be appropriate experimental materials.

2.3.2. Target pictures

Target pictures included 20 aggressive pictures (e.g., knives, guns) and 20 non-aggressive pictures (e.g., flowers, grass). We used Adobe Photoshop 2018 to modify the pictures to ensure that all pictures had a white background, with all objects in similar size and brightness. Target pictures were presented at the center of the computer screen.

2.3.3. Modified semantic classification task

A modified semantic classification task (MSCT) was used to measure the accessibility of aggressive cognitions. Instead of using abstract words as in the traditional SCT (Bowers & Turner, 2003), we used pictures because of children's age. Participants were asked to decide whether the target object shown at the center of the screen was aggressive or non-aggressive. For example, knife, gun and sword were used as aggressive pictures, whereas flower, banana and grass were used as non-aggressive pictures. Before the MSCT, the lab assistants used a cover story to tell the participants that the task was to assess their response speed and accuracy. Participants were asked to press

[1] This is not necessarily the same as high trait aggression, as traits are difficult to measure at this age.

Table 1
Means and standard deviations in judgments of cartoons (N = 61).

Cartoons	Handy Manny M ± SD	Peppa Pig M ± SD	PAW Patrol M ± SD	Boonie Bears M ± SD	Big-Headed Kid and Small-Headed Father M ± SD
Prosocial scene	4.07 ± 0.65	3.39 ± 0.78	3.36 ± 0.90	3.57 ± 1.07	3.90 ± 0.70
Prosocial content	4.07 ± 0.70	3.52 ± 0.70	3.39 ± 0.86	3.61 ± 1.08	3.93 ± 0.70
Interest/Fun	3.59 ± 0.94	4.07 ± 0.85	3.34 ± 0.95	4.21 ± 0.73	3.46 ± 0.89
Popularity	3.59 ± 0.64	4.43 ± 0.62	3.56 ± 0.70	4.26 ± 0.68	3.69 ± 0.67
Excitement	3.07 ± 0.98	3.64 ± 1.02	2.82 ± 0.97	3.85 ± 0.85	3.08 ± 0.95
Pleasure	3.62 ± 0.93	3.98 ± 0.87	3.30 ± 0.92	4.02 ± 0.67	3.54 ± 0.98
Total	3.67 ± 0.58	3.84 ± 0.54	3.30 ± 0.68	3.92 ± 0.60	3.60 ± 0.59

number "1″ on the keyboard if an aggressive picture (e.g., sword) was presented, and press "2" if a non-aggressive picture (e.g., banana) was presented. For each trial, a fixation "+" appeared at the screen center for 300 ms (ms), followed by the target picture for 4000 ms. A blank screen would appear for 100 ms after participants made a response by pressing a button, and the program would enter the next trial. The program automatically entered the next trial if participant did not respond within the given 4000 ms. Participants could quit the task at any time if they felt uncomfortable.

MSCT was divided into practice and formal sessions. The practice session included 12 practice trials that were not included in the formal session. The practice session aims to familiarize the participants with key pressing. The program would not advance to the formal session until the participant's accuracy reached 90%. The formal session included two blocks with 20 trials (10 aggressive pictures and 10 non-aggressive pictures) in each block constituting a total of 40 trials (20 aggressive pictures and 20 non-aggressive pictures). Each picture was presented once in each block. Participants had a short rest for 60 s between the two blocks. The participants' average accuracy rating ranged from 85% to 95% and no missing data was found. The reaction time to aggressive pictures represents aggressive cognitions.

According to the cognitive-neoassociative hypothesis, cognitions can be activated through associative pathways, activating related thoughts along the way (Berkowitz, 1990). Previously existing connected concepts should be more quickly accessible than less related concepts. For example, playing a prosocial (as opposed to a neutral) video game resulted in faster reaction times for identifying prosocial words (Greitemeyer & Osswald, 2011). Thus, longer reaction time is indicative of less accessibility of aggressive cognition, and vice versa for the shorter reaction time.

2.3.4. Competitive reaction time task

The competitive reaction time task (CRTT), programmed in E-prime 3.0, was used to measure aggressive behaviors. The CRTT is widely accepted as a valid measure of aggressive behavior in laboratory settings (Warburton & Bushman, 2019). The noise intensity (50 dB, 60 dB, 70 dB, 80 dB) set by children to "virtual opponents" with the corresponding number buttons (1, 2, 3, 4) represents aggressive behavior. We set the noise intensity level as 50 dB, 60 dB, 70 dB and 80 dB to protect the young children. Higher noise intensity is considered to be more aggressive, whereas lower noise intensity was considered less aggressive. The participants were instructed to compete with "virtual opponents" to see who was the first to respond to the presented tone. The loser would receive a loud noise blast with the intensity set by the opponent after each attempt. The noise intensity set by the participants for "virtual opponents" represented aggressive behaviors when the participants won this trial.

The CRTT consisted of practice and formal sessions from prior research (Anderson & Carnagey, 2009). A practice session included 12 trials, during which participants could randomly hear 4 different levels of noise intensity (50 dB vs. 60 dB vs. 70 dB vs. 80 dB). The formal session included two steps: Step 1 included 25 trials, and the first trial was set to be always a "loss" (e.g., participants would always hear the noise blast regardless of how fast they reacted) to make participants feel frustrated. The remaining 24 trials were then divided into three blocks, with 8 trials in each block. The first and the second block had 4 wins and 4 losses while the third block had 5 wins and 3 losses. The last trial was always a "win". The noise intensity set by the virtual opponents for participants ranged from 50 to 80 dB (weak to strong).

At the end of the step 1, lab assistants reminded participants that in step 2 they could set the noise intensity for the virtual opponents, and that they would not receive any noises from virtual opponents in step 2. As in step 1, step 2 included the same 25 trials, but the roles were reversed: If the participant won, he/she could select one noise intensity (50 dB, 60 dB, 70 dB, 80 dB) to punish the virtual opponent by pressing number button (1, 2, 3, 4). No noise setting (0 dB = no aggressive behavior) was also available.

2.4. Procedure

The experiment was approved by the researchers' University Ethics Committee and was consistent with the 1964 Helsinki declaration. Each participant sat in front of a 15-inch laptop in a big hall at each kindergarten. First, lab assistants illustrated the purposes and precautions of the experiment for parents, teachers and children. Participants' parents provided signed informed consents. Second, participants in the experimental group watched each of the five cartoons for 10 min for 5 consecutive days. Specifically, 44 boys and 43 girls watched prosocial cartoons, 43 boys and 44 girls did not watch any cartoons. Then they completed the MSCT and the CRTT. Participants in the control group completed the MSCT and the CRTT without watching any cartoons. Finally, participants received a gift for their participation. The experiment-control mixed design allowed us to evaluate the prosocial cartoon model effects, using SPSS 21.0 and PROCESS 3.0 macro for analysis of variance (ANOVA) and mediation analysis.

3. Results

3.1. Descriptive statistics of aggressive cognitions and aggressive behaviors

Tables 2 and 3 list the means, standard deviation and cell sample size across 4 conditions. The mean level of aggressive cognitions was lower (longer reaction times) for the experimental group than for the control group (shorter reaction times). Watching prosocial cartoons resulted in longer reaction times when identifying aggressive pictures. In contrast, participants who did not watch the prosocial cartoons showed shorter reaction times. The experimental group also showed less aggressive behaviors than the control group. As for gender, the

Table 2
Means and standard deviations of aggressive cognitions (ms).

Condition	Boys M ± SD	N	Girls M ± SD	N
EG	1405.71 ± 427.26	44	1226.24 ± 343.16	43
CG	1100.38 ± 362.71	43	1208.58 + 413.09	44
Total	1254.80 ± 423.20	87	1217.31 ± 378.03	87

Note. ms = milliseconds; EG = experimental group, CG = control group. Aggressive cognitions = mean reaction time to aggressive pictures.

Table 3
Means and standard deviations of aggressive behaviors (dB).

Condition	Boys $M \pm SD$	N	Girls $M \pm SD$	N
EG	56.17 ± 9.93	44	60.75 ± 8.40	43
CG	71.54 ± 8.84	43	64.00 ± 10.05	44
Total	63.77 ± 12.13	87	62.40 ± 9.36	87

Note. dB = decibel; EG = experimental group, CG = control group. Aggressive behaviors = mean noise level setting to opponents.

boys' mean aggressive cognitions and behaviors were lower than that of girls. Based on these preliminary analyses, we conducted specific analysis of these variables.

3.2. Analysis of variance on aggressive cognitions

To test H1 and H2, we conducted a 2 (prosocial cartoon: watching vs. not watching) X 2 (gender: boys vs. girls) analysis of variance (ANOVA) on aggressive cognitions (reaction time to aggressive pictures), and age was controlled as a covariate. There was a significant main effect of prosocial cartoon on aggressive cognitions [$F(1, 169) = 7.45$, $p = 0.007$, $d = 0.42$, part. $\eta^2 = 0.04$]. Children who watched prosocial cartoons ($M = 1315.81$, $SD = 41.75$) displayed lower aggressive cognitions than those who did not ($M = 1154.64$, $SD = 41.75$). Main effect for gender was not significant [$F(1, 169) = 0.37$, $p = 0.54$, $d = 0.09$, part. $\eta^2 = 0.002$]. There was a significant cartoon X gender interaction [$F(1, 169) = 5.93$, $p = 0.02$, $d = 0.37$, part. $\eta^2 = 0.03$]. Further simple effect analysis showed that boys who watched prosocial cartoon models exhibited lower aggressive cognitions than boys who did not watch [F (1, 169) = 13.34, $p < 0.001$, $d = 0.56$, part. $\eta^2 = 0.07$], whereas no significant group differences in aggressive cognitions were found among girls [F (1, 169) = 0.04, $p = 0.84$, $d = 0.03$, part. $\eta^2 < 0.001$), see Table 4 & Fig. 1.

3.3. ANOVA on aggressive behaviors

To test H1 and H2, we also conducted a 2 X 2 ANOVA on aggressive behaviors (noise intensity setting), and age was controlled as a covariate. There was a significant main effect of prosocial cartoon on aggressive behaviors [$F(1, 169) = 42.97$, $p < 0.001$, $d = 1.00$, part. $\eta^2 = 0.20$]. Children who watched prosocial cartoons ($M = 58.47$, $SD = 1.00$) demonstrated less aggressive behaviors than those who did not ($M = 67.77$, $SD = 1.00$). There was no significant main effect for gender predicting aggressive behaviors [$F(1, 169) = 1.07$, $p = 0.30$, $d = 0.16$, part. $\eta^2 = 0.006$]. There was a significant cartoon X gender interaction [$F(1, 169) = 18.25$, $p < 0.001$, $d = 0.65$, part.$\eta^2 = 0.10$]. Simple effect analysis showed that boys who watched prosocial cartoons displayed less aggressive behaviors than boys who did not [$F(1, 169) = 58.61$, $p < 0.001$, $d = 1.17$, part. $\eta^2 = 0.25$], whereas no significant group differences in aggressive behaviors were found among girls [$F(1, 169) = 2.61$, $p = 0.11$, $d = 0.25$, part. $\eta^2 = 0.01$] (see Table 5 & Fig. 2).

Table 4
Analysis of variance (ANOVA) on aggressive cognitions.

Variables	Mean Square	$F(df_1, df_2)$	Cohen's d	Part. η^2
Cartoon	1129623.48	7.45**(1,169)	0.42	0.04
Gender	56224.67	0.37(1,169)	0.09	0.002
Cartoon X Gender	899795.32	5.93*(1,169)	0.37	0.03

Note. $*p < 0.05$, $**p < 0.01$.

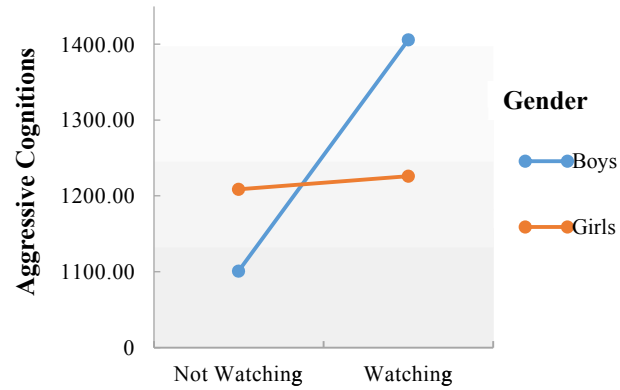

Fig. 1. Interaction Between Prosocial Cartoon and Gender on Aggressive Cognitions. *Note.* Aggressive cognitions represent the average of the reaction time to aggressive pictures (in ms).

Table 5
ANOVA on aggressive behaviors.

Variables	Mean Square	$F(df_1, df_2)$	Cohen's d	part. η^2
Cartoon	3758.71	42.97***(1,169)	1.00	0.20
Gender	93.29	1.07(1,169)	0.16	0.006
Cartoon X Gender	1595.94	18.25***(1,169)	0.65	0.10

Note. $***p < 0.001$.

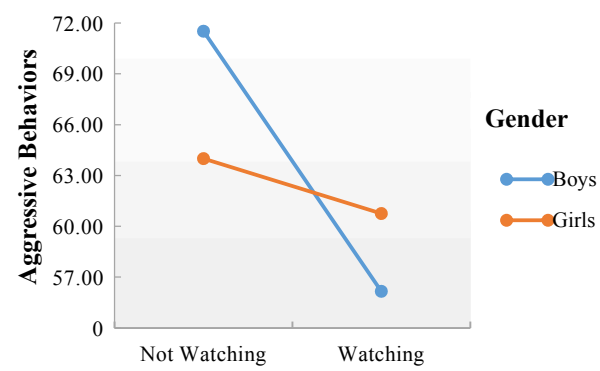

Fig. 2. Interaction Between Prosocial Cartoon and Gender on Aggressive Behaviors. *Note.* Aggressive behaviors represent the average of the noise intensity setting (in dB).

3.4. Aggressive cognitions as a mediator of aggressive behaviors

Based on the ANOVA results that prosocial cartoons decreased both aggressive cognitions and aggressive behaviors, we further tested whether prosocial cartoon viewing decreased aggressive behaviors via aggressive cognitions, as theory predicts. A single mediator model (Mackinnon & Fairchild, 2009) and bootstrapping (5000 bootstrap sample, 95% confidence interval) were used to evaluate the indirect effect of prosocial cartoon viewing on aggressive behaviors (Hayes & Preacher, 2014). We controlled gender as a covariate because of the significant cartoon X gender interaction. All coefficients were standardized (see Fig. 3).

The mediation effect of prosocial cartoon viewing on aggressive behaviors was partially significant ($\beta = -0.14$, $SE = 0.06$, 95% CI [−0.27, −0.03]). The direct effect of prosocial cartoon viewing on aggressive behaviors was significant ($\beta = -0.72$, $SE = 0.13$, 95% CI [−0.98, −0.46]). Prosocial cartoon viewing significantly predicted lower accessibility of aggressive cognitions/longer reaction time ($\beta = 0.40$, $SE = 0.15$, 95% CI [0.11, 0.70]), and lower accessibility of aggressive cognitions/longer reaction time significantly positively predicted lower aggressive behavior ($\beta = -0.34$, $SE = 0.07$, 95% CI

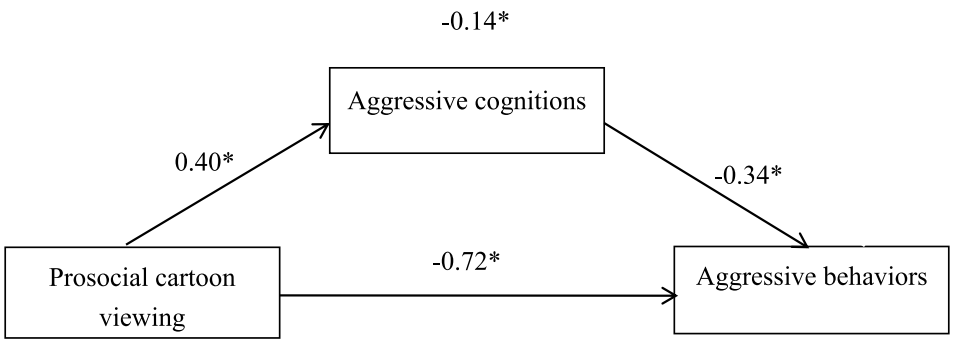

Fig. 3. Partial Mediation Model of Prosocial Cartoon Viewing on Aggressive Behaviors through Aggressive Cognitions. *Note.* 1 = Watching, 0 = Not watching; Standardized coefficients; Solid lines represent significant paths; $^*p < 0.05$.

[−0.47, −0.21]). The total effect of prosocial cartoon viewing on aggressive behavior was significant ($\beta = -0.86$, $SE = 0.14$, 95% CI [−1.13. −0.59]). The direct effect (−0.72) and the mediating effect (−0.14) accounted for 84% and 16% of the total effect (−0.86), respectively. In the present study, both aggressive cognitions and aggressive behaviors decreased for the participants who watched prosocial cartoons. As expected, more prosocial cartoon viewing predicted lower accessibility of aggressive cognitions, and lower aggressive cognitions positively predicted lower aggressive behaviors.

4. Discussion

Consistent with H1, children who watched prosocial cartoons exhibited lower aggressive cognitions and engaged in less aggressive behaviors than those who did not. This finding theoretically supports the GLM, which proposes that exposure to prosocial media changes individuals' cognitions, attitudes, emotions and behaviors (Buckley & Anderson, 2006). This result is also consistent with previous research that prosocial media diminishes aggressive cognitions and behaviors (e.g., Greitemeyer et al., 2012; Greitemeyer & Osswald, 2009; Leeuw, Kleemans, Rozendaal, Anschütz, & Buijzen, 2015; Whitaker & Bushman, 2012). The social learning model suggests that children of this age learn by imitating the cartoon characters' prosocial behaviors (Bandura et al., 1961). Although the results do not contradict the SLM, it is less clear from this theoretical perspective how viewing different types of prosocial behavior would decrease aggressive cognitions. Decreasing aggressive cognition was not modeled in the cartoons, and the control condition simply had no exposure to the aggressive models. In contrast, the GLM proposes that viewing prosocial models changes emotional, motivational, and cognitive states. The GLM seems to be more appropriate in explaining findings of this study and several others (e.g., Coyne et al., 2017; Prot et al., 2014), which demonstrate that prosocial media strengthened the cognitive judgment/performance, which subsequently lowered children's aggression.

Inconsistent with H2, we found that prosocial cartoon models reduced aggressive cognitions and aggressive behaviors more for boys rather than for girls. Interestingly, this finding replicated the past research study where prosocial role model behavior in TV programs increased more helping behavior of boys than those of girls, and boys also aggressed less than girls did (Shuai, 2015). It could be argued that boys have a relatively lower behavioral control capability than girls so that their cognition and behavior were likely to be affected by situational factors (Raaijmakers et al., 2008). On the other hand, boys may be better at adjusting their own cognition and behavior than girls (Cahill, 2014), therefore showing lower aggressive cognitions and behaviors than girls after being exposed to prosocial cartoon models. It also may be that boys have higher base levels of aggressive cognitions and behaviors, and therefore have more room for change. The present study cannot fully answer this question, but it does suggest that boys may be regarded as a key population for reducing aggression by using prosocial cartoons as a strategy.

Consistent with H3, aggressive cognitions functioned as a mediator of later aggressive behaviors. This finding supports the GLM in that the interaction of situational factors and personal factors can affect internal states (i.e., cognition), which in turn influences subsequent behaviors (Gentile et al., 2014). Thus, prosocial cartoon models reduced aggressive behaviors through lowered aggressive cognitions. Although not measured in this study, exposure to prosocial video games can also increase prosocial behavior by increasing prosocial thoughts, which in turn reduces aggression (Greitemeyer & Osswald, 2010).

The present study has several strengths: First, we evaluated the causal effect of prosocial cartoon models on aggressive cognitions and behaviors for kindergartners for five consecutive days, which may have better reliability and validity than previous studies that only tested the immediate effect. Second, to the best of our knowledge, this study is the first experimental study involving young children (i.e., kindergartners) in China, which adds to the diversity and generalizability of the research. This study demonstrated that aggressive cognitions mediate the prosocial cartoon effects on aggressive behaviors, which provides additional insight into prevention and intervention of children's aggression. Finally, by using multiple prosocial cartoon shows, we used stimulus sampling (Wells & Windschitl, 1999), which is also one of the most effective approaches for teaching to transfer (Gentile et al., 2014).

Several limitations should also be noted: First, although the participants in the experimental condition watched prosocial cartoons for five consecutive days, our experiment is not a longitudinal study. Future research should consider collecting longitudinal data to determine the long term effect of prosocial cartoons on young children. Second, although we selected aggressive children nominated by peers and teachers, the sample size of the study was relatively small, and the proportion of gender in each condition was not completely balanced. Thus, this issue may undermine the true effect size. Future research should consider expanding the sample size and balance the gender proportion in each condition to provide stronger conclusions. Third, the control condition used in the study could be a potential concern as children in the control condition did not watch any cartoons. It is possible that the aggression reducing effect of watching the prosocial cartoons has little to with the prosocial content but more with watching any cartoons. For example, it may be that watching the cartoons is a joyful experience and thus aggression is reduced. Therefore, future researchers may use non-prosocial cartoons for the control group. At last, we did not evaluate the specific effects of prosocial cartoon model sub-themes (e.g. sharing, cooperating, donating, helping, comforting) on aggressive cognitions and aggressive behaviors. Future research could consider testing the effect of these specific sub-themes.

Our study has several implications: The study results support the GLM that brief exposure to prosocial media can decrease individuals' aggression, at least partly by changing aggressive cognitions.

Researchers have previously suggested that prosocial media could be used as an intervention to children's aggression (Gentile et al., 2014). This study has some important practical implications on how children's aggression can be decreased. In addition, the mediating effect found in this study reconfirmed that watching prosocial cartoon models reduced aggressive behaviors through decreased aggressive cognitions. Therefore, parents and teachers could decrease children's aggressive behaviors of kindergartners by reducing children's exposure to aggressive media exposure and by increasing their prosocial media exposure.

What elements of prosocial cartoons are effective in reducing aggression? Some studies demonstrate that prosocial scenes and contents in children's cartoon shows are more obvious than shows directed toward older audiences or adults (Smith, Smith, Pieper, Yoo, & Bowden, 2006). For example, Disney's TV shows for children often include prosocial content, and the main characters often perform prosocial behaviors for their family, peers, and even strangers (Padilla-Walker, Coyne, Fraser, & Stockdale, 2013). Cartoon developers, teachers and parents should understand the power of prosocial cartoons, which has been reflected in previous research findings for infants, toddlers, and preschoolers (Kirkorian, Wartella, & Anderson, 2008; Linebarger & Vaala, 2010; Thakkar, Garrison, & Christakis, 2006). However, not all prosocial cartoons can lower aggression. Certain cartoons, such as ones that involve stories of superheroes, where characters "help" others by harming some (sometimes called prosocial aggression) could increase rather than decrease aggression (Liss, Reinhardt, & Fredriksen, 1983). Bearing this in mind, cartoon producers, educators and public policy makers should focus on prosocial attributes of cartoons to diminish children's aggression.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.childyouth.2020.105498.

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Corrigendum

Corrigendum to "Effects of prosocial cartoon models on aggressive cognitions and aggressive behaviors" [Child. Youth Serv. Rev. 118 (2020) 105498]

Qian Zhang [a,b,*], Yi Cao [b,c], JingJin Tian [b,d], Kim El-Lim [e], Douglas A. Gentile [e]

[a] Center for Studies of Education and Psychology of Minorities in Southwest China, Southwest University, Chongqing, China
[b] Faculty of Education, Southwest University, Chongqing, China
[c] The Third Kindergarten in Chengdu, Sichuan, China
[d] The Dawn Innovation Kindergarten, Chongqing, China
[e] Department of Psychology, Iowa State University, Ames, IA, USA

The authors regret some data reporting errors of aggressive cognitions were discovered. To assure the accuracy of the report, the following sections present the new results. The results did not influence the conclusion of the original article.

Descriptive statistics of aggressive cognitions

See Table 2.

Analysis of variance on aggressive cognitions

There was a significant main effect of prosocial cartoon on aggressive cognitions [$F(1, 169) = 5.77$, $p = 0.02$, $d = 0.37$, part. $\eta^2 = 0.03$].

Table 2
Means and standard deviations of aggressive cognitions (ms).

Condition	Boys		Girls	
	$M \pm SD$	N	$M \pm SD$	N
EG	1376.80 ± 359.22	44	1232.78 ± 331.45	43
CG	1154.58 ± 287.18	43	1222.41 + 287.14	44
Total	1266.97 ± 342.47	87	1227.54 ± 308.06	87

Note. ms = milliseconds; EG = experimental group, CG = control group. Aggressive cognitions = mean reaction time to aggressive pictures.

Table 4
Analysis of variance (ANOVA) on aggressive cognitions.

Variables	Mean Square	$F(df_1, df_2)$	Cohen's d	Part. η^2
Cartoon	584242.60	5.77*(1, 169)	0.37	0.03
Gender	64391.22	0.64(1, 169)	0.12	0.004
Cartoon × Gender	488019.61	4.82*(1, 169)	0.34	0.03

Note. $^*p < 0.05$.

Children who watched prosocial cartoons ($M = 1304.60$, $SD = 34.13$) displayed lower aggressive cognitions than those who did not ($M = 1188.69$, $SD = 34.13$). Main effect for gender was not significant [$F(1, 169) = 0.64$, $p = 0.43$, $d = 0.12$, part. $\eta^2 = 0.004$]. There was a significant cartoon × gender interaction [$F(1, 169) = 4.82$, $p = 0.03$, $d = 0.34$, part. $\eta^2 = 0.03$]. Further simple effect analysis showed that boys who watched prosocial cartoon models exhibited lower aggressive cognitions than boys who did not watch [$F(1, 169) = 10.56$, $p = 0.001$, $d = 0.50$, part. $\eta^2 = 0.06$], whereas no significant group differences in aggressive cognitions were found among girls [$F(1, 169) = 0.02$, $p = 0.88$, $d = 0.02$, part. $\eta^2 < 0.001$), see Table 4 and Fig. 1.

Mediating effect of aggressive cognitions: New analysis

The mediation effect of prosocial cartoon viewing on aggressive

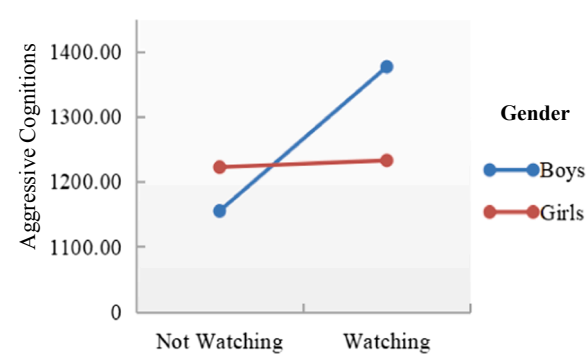

Fig. 1. Interaction between prosocial cartoon and gender on aggressive cognitions. *Note.* Aggressive cognitions represent the average of the reaction time to aggressive pictures (in ms).

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* Corresponding author at: Faculty of Education, Southwest University, Tiansheng Rd. 2, Beibei District, Chongqing 400715, China.
E-mail addresses: zhq@swu.edu.cn (Q. Zhang), ellimkim@iastate.edu (K. El-Lim), dgentile@iastate.edu (D.A. Gentile).

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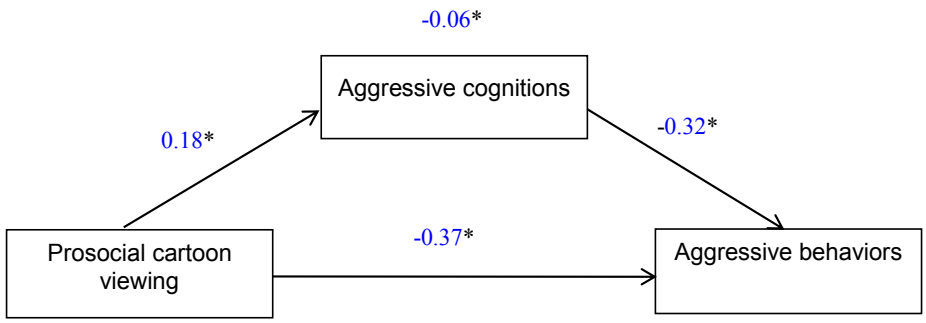

Fig. 3. Partial mediation model of prosocial cartoon viewing on aggressive behaviors through aggressive cognitions. *Note.* 1 = Watching, 0 = Not watching; All standardized coefficients; Solid lines represent significant paths; $^*p < 0.05$.

behaviors was partially significant ($\beta = -0.06$, $SE = 0.03$, 95% CI [−0.11, −0.01]). The direct effect of prosocial cartoon viewing on aggressive behaviors was significant ($\beta = -0.37$, $SE = 0.07$, 95% CI [−0.50, −0.24]). Prosocial cartoon viewing significantly predicted lower accessibility of aggressive cognitions/longer reaction time ($\beta = 0.18$, $SE = 0.08$, 95% CI [0.03, 0.33]), and lower accessibility of aggressive cognitions/longer reaction time significantly positively predicted lower aggressive behavior ($\beta = -0.32$, $SE = 0.07$, 95% CI [−0.45, −0.19]).

See Fig. 3.

The authors would like to apologize for any inconvenience caused.